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SARDAR PATEL UNIVERSITY

SYBCA(SEM-III) On Demand Examination - 2020

US03CBCA23 : OBJECT ORIENTED PROGRAMMING THROUGH C++

Date:11/02/2020

Time: 10.00 AM to 1.00 PM

Total Marks : 70

Q.1 Multiple choice questions:

[10]

- _____ is process by which objects of one class get the properties of other class.
A. Inheritance
B. Polymorphism
C. Data hiding
D. Encapsulation
- _____ is called the insertion or put to operator.
A. &
B. <<
C. >>
D. ::
- C++ supports _____.
A. anywhere declaration of variable
B. Dynamic initialization of variable
C. Reference variable
D. All
- _____ function is used to copy one string into another string.
A. strcpy()
B. stringcpy()
C. strcopy()
D. StrCopy()
- _____ is a variable of type class.
A. Static
B. Object
C. Operator
D. Constant
- _____ access specifier is default in class definition.
A. private
B. protected
C. public
D. None of Above
- One base class & multiple derived classes means _____ inheritance.
(A) multiple
(B) hierarchical
(C) multilevel
(D) hybrid
- In case of multipath inheritance, problem of duplicate data-members is resolved using _____ keyword.
(A) abstract
(B) const
(C) virtual
(D) default
- Which one of the following operators can be overloaded?
A. ::
B. New
C. .*
D. Sizeof
- Which of the following declarations are illegal?
A. Void *ptr
B. Char *str1 = "xyz"
C. Char str2 = "abc"
D. Const *int p1

Q.2 Write short answer (any 10)

[20]

1. Define class & object as concept of OOP.
2. List out the features of OOP.
3. Differentiate variable and constant.
4. Define destructor and give its example.
5. Define nameless and arrays of objects.
6. Explain cin & cout in C++ with example.
7. Discuss default arguments briefly.
8. Define friend function & list its use.
9. What is a virtual base-class? What is the use of it?
10. What is stream?
11. What is Operator Overloading?
12. List out five operators to be overloaded in C++.

Q.3(A) What is C++? Explain structure of C++.

[4]

Q.3(B) Explain features (characteristics) of OOP in detail.

[6]

OR

Q.3(A) Differentiate OOP & POP.

[4]

Q.3(B) Explain different operator available in C++.

[6]

Q.4(A) What do you mean by constructors? What is the use of constructors in C++?

[6]

Q.4(B) Explain declaration and initialization of one dimensional array with example.

[4]

OR

Q.4(A) How you can create arrays of objects? Give an example and explain it.

[6]

Q.4(B) Explain nameless objects with proper example.

[4]

Q.5 Define inheritance. Explain all types of inheritance. Discuss the syntax of defining base-class & derived class.

[10]

OR

Q.5 Discuss inline functions with appropriate example & Also discuss briefly – virtual function.

[10]

Q.6(A) Explain binary operator overloading with example.

[5]

Q.6(B) Explain read(), write(), seek(), close()

[5]

OR

Q.6(A) Explain unary operator overloading with example.

[5]

Q.6(B) What is file mode? Describe the various file mode operations available.

[5]

—X—